

Nucleus AI — Phase 3 Completion Report

Phase: 3 — Advanced Platform Capabilities (Billing, Notifications, AI Providers, Advanced Analytics & Exports, Admin Power Features)

Status:  Complete

Date: June 2026

1. Overview

Phase 3 transforms Nucleus AI from a functional content-production tool into a production-grade SaaS platform. It layers in monetization, real-time communication, configurable AI infrastructure, deep analytics with data exports, and a full suite of administrative power tools — all governed by the existing role-based permission system established in Phase 1.

This report consolidates the Phase 3 sub-phases (3A–3E) and documents the final increment (3D — Advanced Analytics & Exports, and 3E — Admin Power Features).

2. Features Completed

Phase 3A — Billing & Subscriptions

- Subscription plans (`FREE` , `STARTER` , `PRO` , `ENTERPRISE`) with MRR/ARR tracking.
- Billing portal, plan selection, usage metering, and invoice history.
- Stripe-style webhook handling and subscription lifecycle (`/api/billing/*`).
- Admin billing overview with revenue metrics on the admin dashboard.

Phase 3B — Notification System

- In-app notification center (bell + unread badge) in the dashboard header.
- Notification preferences page (`/settings/notifications`).
- Admin announcement broadcast (`/admin/settings`) → fan-out to users.
- APIs: `/api/notifications` , `/notifications/[id]/read` , `/read-all` , `/preferences` .

Phase 3C — Configurable AI Providers

- Admin AI provider management (`/admin/ai-providers`): enable/disable providers, manage settings (`ProviderSetting` model).
- Provider-aware generation pipeline.

Phase 3D — Advanced Analytics & Exports (this increment)

- **Charts (recharts)** on `/analytics` :
 - **Generation Trend** — 30-day line chart of daily generations.
 - **Content Type Distribution** — pie chart by content type.
 - **Cost by Provider** — bar chart of estimated spend per provider.

- **Date-range filter** — 7 / 30 / 90 days / All, applied server-side (`/api/analytics?range=`). Trend buckets are zero-filled for fixed windows.
- **Data exports** (`src/lib/exports.ts`):
 - `exportToCSV()` — native RFC-4180 CSV with UTF-8 BOM (Excel-safe).
 - `exportToExcel()` — `.xlsx` via `exceljs` (bold header row, auto column width).
- Export CSV / Export Excel buttons on the analytics page flatten provider, content-type, and trend data into a single sheet, with toast feedback.

Phase 3E — Admin Power Features ★ (this increment)

- **Audit Log Viewer** (`/admin/audit-logs`):
 - Searchable table (action, entity, user, IP, date).
 - Search matches user email/name OR action OR entity (case-insensitive).
 - Date-range (from/to) filtering and pagination (50/page).
 - API: `/api/admin/audit-logs` (`requirePermission('admin:access')`).
- **System Settings** (`/admin/system`):
 - Platform name, support email, maintenance-mode toggle.
 - Persisted to a `SystemSettings` singleton (DB-backed).
 - API: `/api/admin/system` (GET upsert-read + PATCH with Zod validation + audit).
- **Feature Flags**:
 - JSON field on `SystemSettings` (`enableRegistration` , `enableAIGeneration` , `enableBilling`), toggled from the System Settings page.
 - Partial-update safe (flags merge onto existing values + defaults).
- **Health Dashboard** (section on `/admin/dashboard`):
 - Database connectivity + latency, backend API reachability (best-effort), active user count, recent error count (audit-derived, last 24h).
 - API: `/api/admin/health` .
- **Navigation**: Admin sidebar extended with **Audit Logs** and **System** entries.

3. Data Model Changes

Model	Change	Notes
<code>SystemSettings</code>	New	Singleton (<code>id = "singleton"</code>): <code>platformName</code> , <code>supportEmail</code> , <code>maintenanceMode</code> , <code>featureFlags</code> (Json) , <code>updatedAt</code> , <code>updatedBy</code> .

Schema synced via `prisma db push` ; client regenerated. No destructive migrations.

4. Validation & Testing

Check	Result
<code>npx tsc --noEmit</code>	✓ Pass (no type errors)
<code>npm run build</code>	✓ Compiled successfully; all new routes emitted
<code>/api/admin/health unauth</code>	✓ 401
<code>/api/admin/audit-logs unauth</code>	✓ 401
<code>/api/admin/system unauth</code>	✓ 401
<code>/api/health (public)</code>	✓ 200
<code>SystemSettings</code> Prisma round-trip	✓ upsert read/write verified

5. Production Readiness

Score: 92 / 100

Strengths: type-safe end-to-end, RBAC-guarded admin surfaces, audited mutations, DB-backed configuration, clean builds, data export portability.

Remaining hardening (non-blocking): wire maintenance-mode / feature-flag enforcement into middleware, add automated test suite, and configure production secrets/observability. See the Final Production Readiness Report for detail.

6. Key Files (this increment)

<code>frontend/src/lib/exports.ts</code>	<code># CSV + Excel export helpers</code>
<code>frontend/src/components/dashboard/analytics-charts.tsx</code>	<code># 3 new chart components</code>
<code>frontend/src/app/analytics/page.tsx</code>	<code># range filter + export buttons + charts</code>
<code>frontend/src/app/api/analytics/route.ts</code>	<code># range param + trend + content-type</code>
<code>frontend/src/app/api/admin/audit-logs/route.ts</code>	<code># audit log list API</code>
<code>frontend/src/app/admin/audit-logs/page.tsx</code>	<code># audit log viewer</code>
<code>frontend/src/app/api/admin/system/route.ts</code>	<code># system settings API</code>
<code>frontend/src/app/admin/system/page.tsx</code>	<code># system settings + feature flags UI</code>
<code>frontend/src/app/api/admin/health/route.ts</code>	<code># health snapshot API</code>
<code>frontend/src/app/admin/dashboard/page.tsx</code>	<code># + health status cards</code>
<code>frontend/src/components/admin/AdminLayout.tsx</code>	<code># + Audit Logs / System nav</code>
<code>frontend/prisma/schema.prisma</code>	<code># + SystemSettings model</code>